

	Hence, $\frac{1}{2} \lambda = 60 \text{ mm}$, so $\lambda = 120 \text{ mm}$
19	D Contribution of E-field at the centre by a pair of like charges opposite each other will cancel out but by a pair of unlike charges, it will result in an E-field strength of 2 units pointing towards the electron. Hence, the net E-field for A is $2\sqrt{2}$ units, 225° (3^{rd} quadrant); B is 2 units, 45° (1^{st} quadrant); C is 2 units, 270°; D is less than 2 units, 112.5° (2^{nd} quadrant).
20	C